Stage 16N Report

Updated 2026-06-22

1 Overview

Stage 16N investigates restart-preserved cycle-jump behavior for the Chaboche UMAT Abaqus benchmark. The R4H diagnostic tested whether source-split restart files differ from restart records produced by long restarted replay runs. The R4I-R recovery rerun then repeated the cycle-280 source-split branch with durable lightweight copy-back and absolute reference CSVs.

2 Results

R4H completed six PBS/Abaqus jobs successfully with `Exit_status=0`. Long-replay restart records passed exactly: R4H1 at cycle 500, R4H3 at cycle 500, and R4H5 at cycle 750 all had zero global, local, and diagnostic S11 error. Short source-split restart cases remained inconsistent: R4H2 failed at cycle 500 with 11.83033% max primary local error, R4H4 was review-level at cycle 500 with 9.3860617%, and R4H6 was review-level at cycle 750 with 8.0369449%.

The R4I-R first wave finished by 2026-06-20 13:28 CEST and was absent from the live PBS queue on 2026-06-21 05:47 CEST. PBS history no longer returned records for jobs `1350601` and `1350602`, so the current classification uses scratch case status files, PBS stdout, Abaqus completion messages, and copied comparison CSVs. R4I-R1 passed exactly at cycle 500; R4I-R2 failed with the same source-split signature as R4H2.

Case	Mode	Endpoint	Status	Global %	Primary local %	S11 %
R4H1	Long replay 280–500	500	pass	0	0	0
R4H2	Source 250–281, restart 280–500	500	fail	1.1887403	11.83033	0.92936729
R4H3	Long replay 270–500	500	pass	0	0	0
R4H4	Source 250–271, restart 270–500	500	review	2.5314199	9.3860617	1.2392968
R4H5	Long replay 505–750	750	pass	0	0	0
R4H6	Source 500–506, restart 505–750	750	review	0.31896796	8.0369449	1.5702038

R4I-R first-wave recovery

Case	Source solve	Restart	Endpoint	Status	Global %	Primary local %	S11 %
R4I-R1	Deck-clone 250–281	280–500	500	pass	0	0	0
R4I-R2	Buffered 250–300	280–500	500	fail	1.1887403	11.83033	0.92936729

R4I-R1 shows that cloning/truncating the clean direct replay deck shape repairs the cycle-280 branch. R4I-R2 shows that a longer buffered generated source alone does not repair that branch; the dominant mismatch remains `HOLE_RING_SDV8_MAX`.

R4I-R second wave and deck-clone confirmations

R4I-R3 and R4I-R4 were submitted on 2026-06-21 through the guarded storage-gated scratch workflow. Both jobs requested one node with 16 cores, one MPI rank, 16 OpenMP threads, 90 GB memory, and 24 h walltime. R4I-R3 reproduced the generated-buffer review signature at restart 270 with 9.3860617% maximum primary local error. R4I-R4 reproduced the generated-buffer review signature at restart 505 with 8.0369449% maximum primary local error.

R4I-R5 then confirmed the deck-clone/truncate strategy for the 270–500 branch: deck-clone source 250–271, restart 270–500, zero global, primary-local, and S11 error at cycle 500. R4I-R6 and R4I-R6B did not classify the 505–750 branch scientifically. R4I-R6 failed during continuation `.stt` writing near cycle 731, and R4I-R6B failed earlier during source `.stt` writing near cycle 503. These are infrastructure/output-writing failures, not numerical method failures.

R4K controller preparation

A 2026-06-22 storage/source audit showed `/scratch9` at 99% use and found no visible clean cycle-505 restart source under `/scratch9/pr21vyci`, `/scratch/pr21vyci`, or the home project clone. The next prepared work is therefore split into two controllers: R4K1 runs the confirmed 250-branch exact deck-clone control `250 -> exact 280 -> 500`; R4K2 performs a 505-branch restart-source preflight and stops before Abaqus solve if no valid source is found. Both controllers include phase timing and lightweight evidence copy-back.

The controllers were submitted on 2026-06-22 through the guarded wrapper. R4K1, job `1350764.mmaster02`, finished with `Exit_status=0` after 06:01:15 walltime and 20:39:54 CPU time. Abaqus completed successfully, and the exact-control comparison at cycle 500 passed with zero maximum global error, zero maximum primary-local error, and zero diagnostic S11 error. This confirms the 250-branch deck-clone exact control `250 -> exact 280 -> 500`. The corrected R4K2 preflight is job `1350767.mmaster02`; it finished successfully in 6 s walltime and found one large non-R4I candidate from the earlier R4E exact-target scratch area, `stage16n_r4e2_exact_500_to_505_solve_506_to_750_exact_target_source`, with a 4.9 GB `.stt` and successful cycle-505 source `.sta`.

The follow-up storage-light validation controller R4K2B, job `1352353.mmaster02`, used that preserved R4E2 cycle-505 candidate directly. It did not generate a new source `.stt`; the scratch case linked the preserved source files, read `STEP=505, INC=54`, and continued cycles 506–750 with no continuation restart-write request. PBS recorded `Exit_status=0`, `Stageout_status=1`, 05:41:43 walltime, 19:16:27 CPU time, and about 3.38 average active cores out of 16. Abaqus completed through cycle 750, but the comparison was nonzero: 0.31896796% maximum global error, 8.0369449% maximum primary-local error, and 1.5702038% diagnostic S11 error. Therefore R4K2B is a scientific validation failure of the R4E2 505 candidate, not an infrastructure failure. The stage-out warning is recorded separately because lightweight evidence was recovered and committed.

The 250-cycle restart branch is now validated for deck-clone/truncate source construction, including the exact-control case R4K1 with zero global and local comparison error. In contrast, the 505-cycle branch remains unresolved. The preserved R4E2 cycle-505 restart candidate completed the R4K2B continuation but reproduced the previous review signature, with 8.0369449% maximum primary-local error. Therefore, the R4E2 candidate is not accepted as a validated 505 -¿ 750 continuation source, and further 505-branch work is postponed to avoid additional large restart-file generation. The next work should focus on storage-light true-jump candidates on the validated 250 branch.

R4L prepared storage-light true-jump controller

The next prepared production controller is `stage16n_r4l_250branch_storage_light_controller`. It uses the validated 250-branch deck-clone/truncate source construction and keeps the continuation decks restart-read-only. The controller runs R4L1 first, targeting the conservative 250 -¿ 270 -¿ 500 true-jump case, then runs R4L2 around 250 -¿ 280 -¿ 500 only if R4L1 passes. If R4L1 reviews or fails, R4L2 is not run; instead the controller writes diagnostic summaries for source-cycle state, first solved continuation cycle, comparison details, and local hole-ring metric ranking. R4L copies only lightweight evidence and deletes case-local heavy Abaqus files after classification.

After confirming no active jobs and `/scratch9/pr21vyci` at about 2.7 TB, R4L was submitted twice. Job `1353907.mmaster02` stopped before Abaqus solve because the first source linker expected a retained R1A `.odb`. The corrected job `1353908.mmaster02` used a cached 270 jump state but still stopped before the R4L1 source solve because Abaqus/Standard restart requires `res`, `prt`, `mdl`, `stt`, and `sim`, and the retained R1A `.mdl` is missing. Therefore R4L is not scientifically classified; the current blocker is incomplete retained R1A restart provenance after storage cleanup.

A restart-source inventory was then run without submitting any solver job. It found no complete retained heavy source set for R4I-R1, R4I-R5, or R4K1. R1A remains incomplete because its `.mdl` is missing or broken. R1B is the only viable 250-branch restart candidate found: `.stt`, `.res`, `.mdl`, `.prt`, `.sim`, and `.sta` are accessible, and the `.sta` completed cleanly at cycle 250. Its `.odb` is missing or broken, so an R4L redesign using R1B must avoid ODB-dependent state extraction and use cached validated jump-state files or another lightweight state source.

The R4L redesign has therefore been prepared as `stage16n_r4l2_250branch_r1b_restart_storage_light_controller`. The controller uses R1B directly as `oldjob=stage16n_r1b_restart_ref_250cycles`, avoids R1A completely, and does not require the broken/missing R1B `.odb`. The 2026-06-22 pre-submit gate found no active jobs, `/scratch9/pr21vyci` at about 2.7 TB, and readable R1B `.stt/.res/.mdl/.prt/.sim/.sta` companions; the R1B `.sta` completed successfully at cycle 250 with restart row `STEP=250, INC=58`. R4L2-1 tests the target-270 true jump and solves 271–500. R4L2-2 tests target 280 and solves 281–500 only if R4L2-1 passes. If cached jump-state files are absent, the controller records `blocked_missing_cached_jump_state` and stops before Abaqus.

The first R4L2 submission, job `1353909.mmaster02`, is not a scientific true-jump result. PBS finished with `Exit_status=0` and `Stageout_status=1` after 00:01:17 walltime and 00:00:17 CPU time. R1B preflight passed and cached target-270 jump state was found, but R4L2-1 failed in the Abaqus input file processor during datacheck and again during solve before a valid continuation/comparison. R4L2-2 was not run. The runner has been patched after this attempt to stop immediately on datacheck/solve failure and retain small `.dat` tail diagnostics before heavy cleanup.

Case	PBS job	Endpoint	Status	Global %	Primary local %	S11 %
R4K1	1350764	500	pass	0	0	0
R4K2B	1352353	750	review	0.31896796	8.0369449	1.5702038

3 Images

No R4H images were generated in this update.

4 Tables

The key R4H and R4I-R comparison tables are included in the Results section. Lightweight R4I-R evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4ir_restart_source_buffer_recovery/
```

Prepared R4K controller files are stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4k_deck_clone_controllers/
```

R4K1 lightweight result evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4k_deck_clone_controllers/
R4K1_250branch_controller/
```

R4K2B lightweight validation evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4k_deck_clone_controllers/
R4K2B_505candidate_validation_controller/
```

Prepared R4L controller files are stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4l_250branch_storage_light_controller/
R4L_250branch_storage_light_controller/
```

The R4L setup-failure result note is stored at:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4l_250branch_storage_light_controller/
STAGE16N_R4L_RESULT.md
```

Prepared R4L2 R1B restart controller files are stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4l2_250branch_r1b_restart_storage_light_controller/
R4L2_250branch_R1B_restart_storage_light_controller/
```

The R4L2 result note is stored at:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4l2_250branch_r1b_restart_storage_light_controller/
STAGE16N_R4L2_RESULT.md
```

The restart-source inventory is stored at:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/STAGE16N_RESTART_SOURCE_INVENTORY_2026_06_22.md
```

The earlier R4H CSV evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4h_restart_source_diagnostics/
```

5 Failures and Debugging

No R4H job failed at PBS/Abaqus level. The original R4I comparison failed because remote reference CSV paths were missing after extraction, and its scratch outputs were not recoverable. R4I-R fixed that workflow by staging absolute reference CSVs and copying lightweight evidence before comparison. The scientific failure now narrows to generated source-split consistency: R4I-R2 reproduces the R4H2 error even with a larger 250–300 source buffer, and R4I-R3/R4I-R4 show that generated post-target buffering does not repair the 270 or 505 branches. R4I-R6/R4I-R6B additionally exposed scratch `.stt` write failures on the 505 branch.

6 Conclusion and Next Steps

R4H, R4I-R, and R4K1 show that long-replay/deck-clone restart records are clean on the confirmed 250 branch, while generated short-source decks remain non-equivalent even with post-target buffering. Deck-clone/truncate is confirmed for 250–270–500 and for the exact-control path 250–280–500. The 505–750 branch now has a completed R4K2B validation of the preserved R4E2 cycle-505 candidate, and that candidate failed scientifically with the same nonzero 750-cycle signature observed in earlier source-split/review cases. R4J9/R4J10 remain blocked. The next useful solver path is the prepared R4L storage-light 250-branch true-jump controller; the 505 branch is parked until a new exact-control gate or redesign is defined.